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RESEARCH INTERESTS

Low-dimensional topology, in particular braids, knots and links, cobordisms, knot concordance, knotted surfaces, Khovanov and Heegaard Floer homology, 3- and 4-manifolds.

EMPLOYMENT

12/2025 – 11/2027 **Postdoctoral fellow** at the [University of Glasgow](#), Glasgow, UK.
Supported by SNSF Postdoc.Mobility fellowship 230329.

10/2023 – 11/2025 **Postdoctoral fellow** at the [Max Planck Institute for Mathematics](#), Bonn, Germany. Mentor: Arunima Ray.

07/2019 – 07/2023 **Scientific assistant** at [ETH Zurich](#), Zurich, Switzerland.

10/2018 – 04/2019 **Teaching assistant** at [Ulm University](#), Ulm, Germany.

08/2017 – 05/2018 **Teaching assistant** at [Syracuse University](#), Syracuse, NY, USA.

10/2013 – 07/2017 **Teaching assistant** at [Ulm University](#), Ulm, Germany.

EDUCATION

07/2019 – 06/2023 **Doctor of Science** in Mathematics at [ETH Zurich](#), Zurich, Switzerland.
Thesis: *On notions of braid positivity and knot concordance*.
Advisor: Peter Feller.

08/2016 – 02/2019 **Master of Science** in Mathematics (1.0, with distinction) at [Ulm University](#), Ulm, Germany. Minor: Computer Science.
Thesis: *Massey products and linking numbers*. Supervisor: Stefan Wewers.

08/2017 – 05/2018 **Master of Science** in Mathematics (GPA 4.0) at [Syracuse University](#), Syracuse, NY, USA.

10/2012 – 08/2016 **Bachelor of Science** in Mathematics (1.5) at [Ulm University](#), Ulm, Germany. Minor: Physics.
Thesis: *Das Existenzproblem von Hurwitz*. Supervisor: Stefan Wewers.

09/2014 – 02/2015 **Visiting semester** at [ETH Zurich](#), Zurich, Switzerland.

SELECTED AWARDS

12/2025 – 11/2027 **Swiss National Science Foundation Postdoc.Mobility fellowship**.
[Project 230329: Concordance of braid positive knots](#).

2017 **Fulbright Travel Grant**.

2014 – 2017 **Talanx Scholarship**.

2013 – 2014 **Deutschlandstipendium** Scholarship (Ulm University, EY Foundation).

2012 – 2013 **Deutschlandstipendium** Scholarship (Ulm University, Rotary Club Ulm).

RESEARCH SEMESTER

Participant in the **AIM Online Research Community** *4-Dimensional Topology* organized by Miriam Kuzbary, Maggie Miller, Juanita Pinzón-Cañedo and Hannah Schwartz in Spring 2021.

PREPRINTS

3. **Braid positive surgery descriptions**, with M. Kegel. [ArXiv:2512.14589](#).
2. **On the detection of knotted spheres by their traces in high dimensions**, with V. Bais, A. Di Prisa, D. Hartman, C.-S. Hsueh, M. Kegel, A. Merz, M. Pencovitch, A. Ray, D. Santoro and L. Wakelin. [ArXiv:2511.07251](#).
1. **Strongly quasipositive links are concordant to infinitely many strongly quasipositive links**. [ArXiv:2210.06612](#).

PUBLISHED AND ACCEPTED PAPERS

6. **Non-complex cobordisms between quasipositive knots**, with M. Borodzik.
Accepted for publication in *J. Math. Pures Appl.* (2025), DOI. [ArXiv:2504.04894](#).
5. **Algorithms in 4-manifold topology**, with S. Bastl, R. Burke, R. Chatterjee, S. Dey, A. Durst, S. Friedl, D. Galvin, A. García Rivas, T. Hirsch, C. Hobohm, C.-S. Hsueh, M. Kegel, F. Kern, S. M. S. Lee, C. Löb, N. Manikandan, L. Mousseau, L. Munser, M. Pencovitch, P. Perrass, M. Powell, J. P. Quintanilha, L. Schambeck, D. Suchodoll, M. Tancer, A. Thiele, M. Uschold, S. Veselá, M. Weiß and M. von Wunsch-Rolshoven.
Accepted for publication in *Algebr. Geom. Topol.*. [ArXiv:2411.08775](#).
4. **Minimal cobordisms between thin and thick torus knots**, with S. Baader, L. Lewark and F. Misev. *Proc. Edinb. Math. Soc., II. Ser.* 69, no. 1 (2026), pp. 177–183.
3. **On the nonorientable four-ball genus of torus knots**, with F. Binns, S. Kang and J. Simone. *Algebr. Geom. Topol.* 25-4 (2025), pp. 2209–2251.
2. **3-braid knots with maximal 4-genus**, with S. Baader, L. Lewark and F. Misev. *Trans. Amer. Math. Soc. Ser. B* 11 (2024), pp. 600–621.
1. **The upsilon invariant at 1 of 3-braid knots**.
Algebr. Geom. Topol. 23-8 (2023), pp. 3763–3804.

CONFERENCE PROCEEDINGS (NOT PEER-REVIEWED)

5. **MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology, Combinatorics, Geometry, and Computation**, with S. Friedl, J. Licata, S. Tillmann. *Oberwolfach Rep.* 21 (2024), no. 3, pp. 2511–2550.
4. **Obstructing ribbon clasp number one**, with D. Celoria, A. Di Prisa, J. Howie, J. Licata, F. Misev, L. Schambeck, L. Wakelin. *Oberwolfach Rep.* 21 (2024), no. 3, pp. 2511–2550.
3. **Notions of braid positivity and knot concordance**. *Oberwolfach Rep.* 21 (2024), no. 3, pp. 2511–2550.
2. **Strongly quasipositive knots are concordant to infinitely many strongly quasipositive knots**. *Oberwolfach Rep.* 20 (2023), no. 1, pp. 172–174.
1. **Concordance of positive braid knots**. *Oberwolfach Rep.* 18 (2021), no. 3, pp. 2360–2362.

INVITED TALKS AT CONFERENCES AND WORKSHOPS

10/2025	Trisections and related topics, Marseille.
11/2024	IberoSing International Workshop 2024: Low-dimensional Topology & Singularity Theory, Madrid.
07/2024	Mini-symposium <i>Knot theory and low dimensional manifolds</i> , European Congress of Mathematics, Seville.
04/2024	NRW Topology Meeting, Wuppertal.
04/2024	Manifolds and groups in Bologna II, Bologna.
09/2023	Leaning into Topology Workshop, Pisa.
09/2023	Swiss Knots, Regensburg.
06/2023	Mini Workshop, Bern.
03/2023	The low dimensional workshop, Budapest,  .
11/2022	Paroles aux jeunes chercheuses et chercheurs en géométrie et dynamique, Orsay.
09/2022	Minisymposium <i>Algebra and Low-Dimensional Topology</i> , DMV Annual Meeting 2022, Berlin.
06/2022	Surfaces in 4-manifolds, Le Croisic.

INVITED TALKS AT SEMINARS (NON-LOCAL)

01/2026	Geometry Seminar, MPI MiS Leipzig.
10/2025	Algebra and Geometry Seminar, Neuchâtel.
07/2025	Oberseminar Global Analysis and Geometry, Regensburg.
06/2025	Séminaire Géométrie des Espaces Singuliers, Lille.
05/2025	Geometry seminar, ETH Zurich.
12/2024	Oberseminar Topologie, University of Bonn.
09/2024	Séminaire de Géométrie, Algèbre, Dynamique et Topologie, Dijon.
05/2024	Séminaire de Géométrie et Topologie de Marseille, Marseille.
05/2024	Oberseminar Topologie, Münster.
04/2024	Geometry & Topology Seminar, Glasgow.
03/2024	MPIM Oberseminar, MPIM Bonn.
11/2023	MPIM Topology Seminar, MPIM Bonn, short talk.
05/2023	Low-dimensional topology seminar, MPIM Bonn.
12/2022	Geometry and Topology Seminar, CIRGET, Montréal, online,  .
06/2022	LKS-Seminar, Regensburg.
05/2022	Geometry & Topology Seminar, Caltech, online.
05/2022	Oberseminar Geometrie, Fribourg.
02/2022	Knot Online Seminar (K-OS), online,  .
04/2021	AIM Online Research Community 4-Dimensional Topology, lightning talk.

INVITED TALKS AT GRADUATE COLLOQUIA

12/2023	Pitt AWM Student Seminar, Pittsburgh, online.
10/2023	Geometry Graduate Colloquium, ETH Zurich.
10/2022	Bern–Fribourg Graduate Seminar, Bern.
05/2022	Baby Geometri Seminar, Pisa.
04/2022	Bernoullis Tafelrunde, Basel.
11/2021	Zurich Graduate Colloquium, Zurich.
04/2021	Geometric Topology Grad and Postdoc Seminar (GT GAPS), online.

CONTRIBUTED TALKS AT CONFERENCES AND WORKSHOPS

02/2024	BTW4: Knot Floer Homology , Besse.
02/2023	Winter Braids XII , Tours.
01/2023	Low dimensional topology , MFO Oberwolfach.
05/2022	Swiss-Mathematical-Society Doctoral Day 2022 , Fribourg.
04/2022	BTW2: Sliceness, exotic pairs, and quantum invariants , Aussois.
09/2021	Invariants and Structures in Low-Dimensional Topology , MFO Oberwolfach, short talk.

OTHER CONFERENCES AND WORKSHOPS ATTENDED

12/2025	Workshop on surfaces in 4-manifolds , Regensburg.
10/2025	36. NRW Topology Meeting , Bonn.
07/2025	Categorification in Low Dimensional Topology , Bochum.
06/2025	Swiss Knots , Geneva.
09/2024	Workshop on 4-manifolds and algorithms , Regensburg.
02/2024	Winter Braids XIII , Montpellier.
09/2023	GRAZP: Groups and Rigidity Around the Zimmer Program , Ventotene.
07/2023	Gauge Theory and Topology: in Celebration of Peter Kronheimer's 60th Birthday , Oxford.
01/2023	Winter school on singularities and low-dimensional topology , Budapest.
08/2022	Frontiers in Geometry and Topology Summer School , Trieste.
04/2022	Braids in Low-Dimensional Topology , ICERM, Providence.
08/2021	Perspectives on quantum link homology theories , Universität Regensburg.
07/2021	Tech Topology Summer School , Georgia Institute of Technology, online.
07/2021	Swiss Knots , Fribourg.
06/2021	NCNGT: Nearly Carbon Neutral Geometric Topology Conference , online.
06/2020	NCNGT: Nearly Carbon Neutral Geometric Topology Conference , online.
02/2020	Winter Braids X , Pisa.
10/2019	Regional workshop on foliations and 3-manifolds , Universität Regensburg.
10/2019	Low-dimensional topology workshop , Universität Regensburg.
09/2019	Workshop on 4-manifolds , Max Planck Institute for Mathematics Bonn.
09/2019	Of coarse! Quasi-isometries and groups: rigidity and classification , Ventotene.
07/2019	Swiss Knots , ETH Zurich.
10/2017	27th annual <i>Route 81</i> conference on commutative algebra and algebraic geometry, Cornell University, Ithaca, NY, USA.

OTHER SHORT RESEARCH STAYS (WITHOUT TALK, CONFERENCE OR WORKSHOP)

07/2025	three days, Humboldt University Berlin.
08/2024	one week, ETH Zurich.
03/2024	three days, ETH Zurich.
04/2022	one week, ICERM, Providence.
09/2020	three days, University of Regensburg.

SUPERVISION

- 02/2025 – 10/2025 **Master thesis** by Chiara Donatone at Università degli Studi di Milano.
Thesis title: *On RBG links and knot traces*.
Co-supervision with Diego Matessi.
- 06/2024 – 06/2025 **Master thesis** by Frieda Kern at the University of Bonn.
Thesis title: *Linking number obstructions to satellite homomorphisms*.
Co-supervision with Peter Teichner. Continues as PhD student at the University of Glasgow.

TEACHING

- University of Bonn/MPIM Bonn**
- Winter 2024/25 Lecturer for **Topics in 3-manifold topology** (Master in Mathematics).
- ETH Zurich**
- Spring 2022 Course organizer for **Analysis II** (Bachelor in Civil, Geospatial, or Environmental Engineering).
- Fall 2021 Course organizer for **Analysis I** (Bachelor in Civil, Geospatial, or Environmental Engineering).
- Spring 2021 Course organizer for **Topologie** (Bachelor in Mathematics).
- Fall 2020 Course organizer for **Lineare Algebra I** (Bachelor in Mathematics, Physics, or Interdisciplinary Sciences).
- Spring 2020 Course organizer for **Algebraic Topology II** (Bachelor or Master in Mathematics).
- Fall 2019 Teaching assistant for **Complex Analysis** (Bachelor in Mathematics, Physics, or Interdisciplinary Sciences).
- Syracuse University**
- Spring 2018 Teaching assistant for **Calculus III**.
- Fall 2017 Teaching assistant for **Calculus III**.
- Ulm University**
- Summer 2019 Instructor for **Linear Algebra Review Course** (Bachelor in Computer Science and Engineering).
- Winter 2013/14 – Summer 2019 Student teaching assistant for **Analysis I-III** and **Linear Algebra I-II**.

TEACHER TRAINING

- Winter 2024/25 Participation in the **Teaching Mentoring Program** at the Hausdorff School for Mathematics in Bonn.
Mentors: Viktoriya Ozornova and Arunima Ray (both MPIM Bonn).
- 09/2021 Certificate **Learning to Teach**.
Program for doctoral teaching assistants at ETH Zurich.
Teaching project: *How to teach writing proofs—classroom assessment techniques in mathematics classes*.

OUTREACH ACTIVITIES

03/2026 (planned)	Co-organizer of GROW@Dresden 2026 (Graduate Research Opportunities for Women at Dresden 2026).
03/2025	Invited panelist for panel discussion “ <i>What is mathematics research like?</i> ” at GROW@Frankfurt 2025 .
12/2024	Invited speaker at Tea Time with Women in Mathematics , Bonn.
04/2024	Volunteer at the mentoring lunch at GROW@Bonn 2024 .
03/2023	<i>Work as a PhD Student</i> , short talk, MindPhair 2023 at ETH Zurich (job-fair for mathematicians, physicists and computational scientists).
03/2022	<i>Work as a PhD Student</i> , short talk, MindPhair 2022 at ETH Zurich.

CONFERENCE AND SEMINAR ORGANIZATION

10/2023 – 11/2025	Low-dimensional topology seminar , MPIM Bonn.
09/2024	MATRIX-MFO Tandem Workshop: Invariants in Low-Dimensional Topology. Combinatorics, Geometry, and Computation at Mathematisches Forschungsinstitut Oberwolfach, Sept 22–27, 2024 (joint with Stefan Friedl, Joan Licata and Stephan Tillmann).
09/2020 – 07/2023	Geometry Graduate Colloquium at ETH Zurich (funded by the Zurich Graduate School in Mathematics).

REFEREEING AND REVIEWING

2024 – present	Referee and quick opinions for <i>Algebr. Geom. Topol.</i> , <i>Bull. Lond. Math. Soc.</i> and <i>Pac. J. Math.</i>
2024 – present	Reviewer for <i>zbMATH Open</i> .

OTHER ACADEMIC SERVICE AND ACTIVITIES

MPIM Bonn	
09/2024	Member of the Postdoc Representative Selection Committee.
ETH Zurich	
09/2021 – 07/2023	Co-organizer of get-togethers for women in math at ETH Zurich and the University of Zurich (funded by the Zurich Graduate School in Mathematics).
08/2021 – 07/2023	Elected member of the Nachdiplom Lectures Committee at the Department of Mathematics.
05/2020 – 05/2023	Member of the Board of VMM (Association of the Mid-level Academic Faculty at the Department of Mathematics).
10/2022	Co-organizer of the first D-MATH (Post)Doctoral Welcome Day.
05/2021 – 05/2022	Mid-level faculty representative in the Department Conference.
Ulm University	
10/2016 – 07/2017	Member of the Mathematics Student Council (Fachschaft Mathematik) at the Department of Mathematics and Economics.

REFERENCES

[Peter Feller](#), University of Neuchâtel,

[Matthew Hedden](#), Michigan State University,

[Lukas Lewark](#), ETH Zurich,

[Viktoriya Ozornova](#), MPIM Bonn (Teaching),

[Arunima Ray](#), University of Melbourne,

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